VISION-LANGUAGE MODELS

Lecture 2

Contrastive Vision-Language Pretraining: CLIP and Related Models

Outline

- **We have covered:**
  - A ViT that turns any image into a sequence of patch tokens — and a single vector via [CLS]/pooling
  - Noted that that transformer encoders are modality-agnostic
- **Now we wish to combine images and text:**
  - An image encoder and a text encoder compute different embeddings. How can we encode a photo of a dog and the sentence “a photo of a dog” so they are comparable?

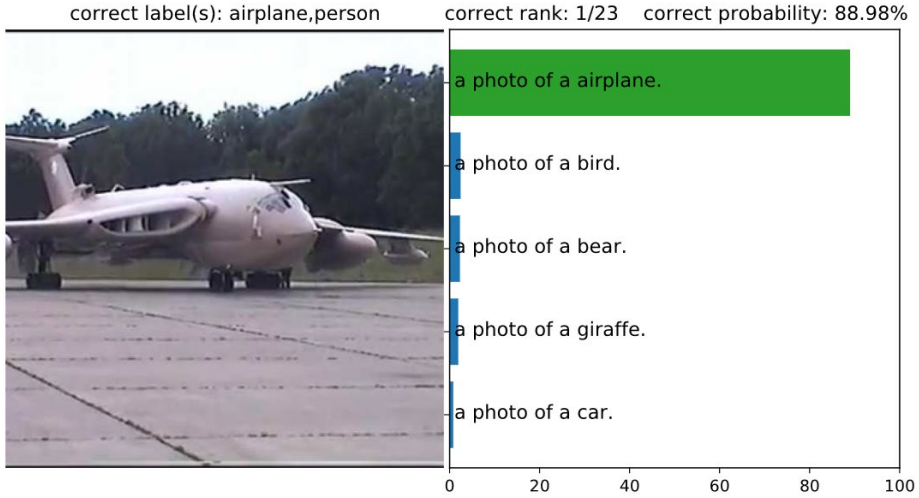
I	**The alignment problem**	Two encoders, one target space — what “aligned” means geometrically
II	**The contrastive objective**	Batches as self-labeling data; the N×N similarity matrix; symmetric InfoNCE loss; temperature
III	**Zero-shot transfer**	Classes as captions; prompt engineering; retrieval; robustness
IV	**Successors and limitations**	ALIGN (scale), SigLIP (sigmoid loss); bag-of-words failures

CLIP: overcoming lack of labels

- **Training via supervision is not viable at scale**
  - ImageNet: 1.3M images, 1000 classes — ~25,000 annotator-years of curation effort behind it
  - A fixed label set is very limited: the model can only map to its 1000 class labels
- **BUT the web is full of free "supervision"**
  - "Alt-text", caption, are (usually) human image annotations — noisy, but very large
  - CLIP approach: 400M (image, text) pairs scraped from the web (the "WIT" dataset) can replace curated labels
- **Natural-language supervision gives us**
  - Open vocabulary: anything describable in English is a potential "class"
  - Richer signal than a class id: "a golden retriever puppy asleep on a red sofa" vs class #207

CLIP: Contrastive Language-Image Pretraining

- **Pretraining: train text and image encoders using all image/text pairs using contrastive loss**
- **Enables building a simple text to image retrieval pipeline using pretrained model**

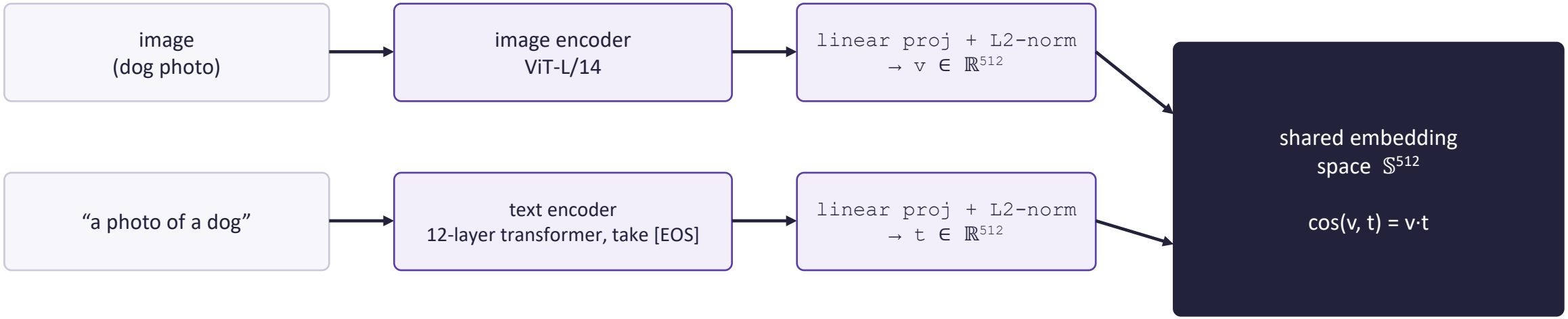


Radford et al. 2021, "Learning Transferable Visual Models From Natural Language Supervision" (arXiv:2103.00020).

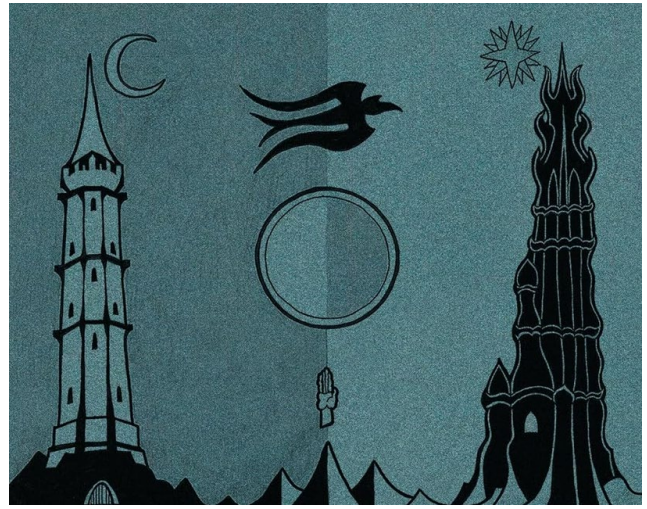
PART I

The alignment problem

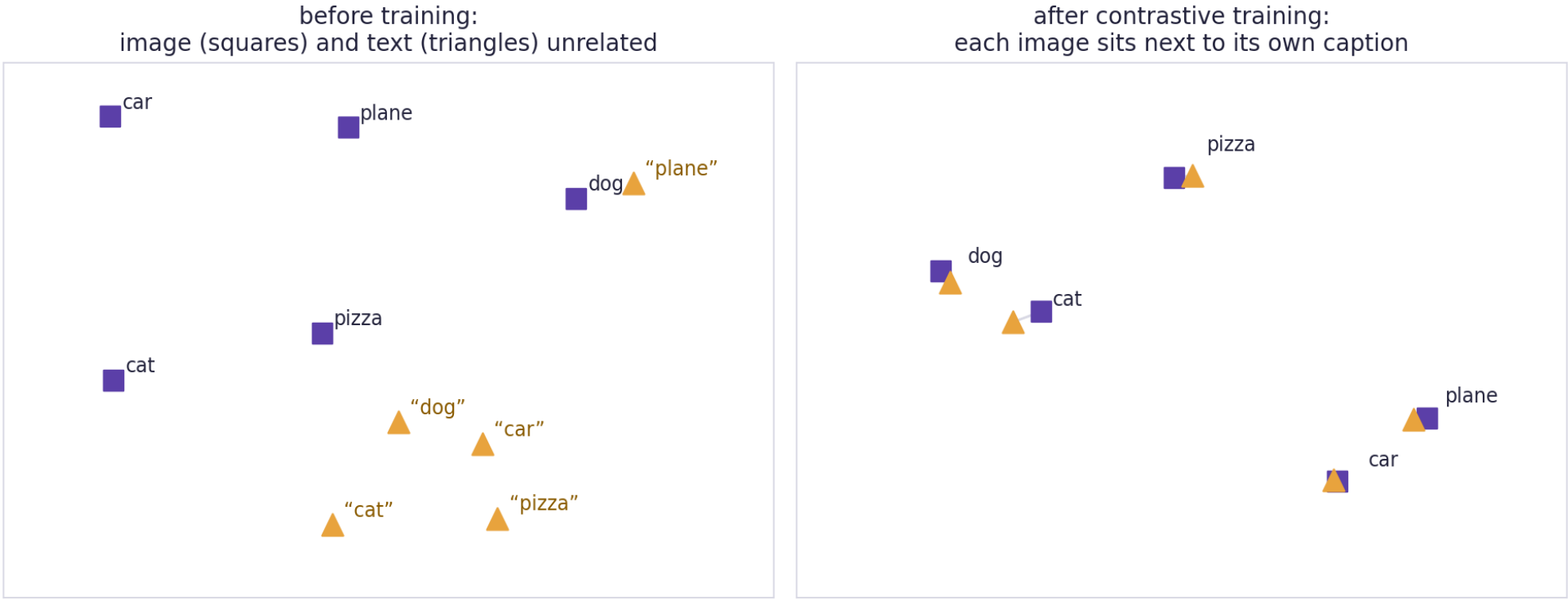
Architecture: the “two-tower” (dual-encoder) design



- Both outputs are L2-normalized → embeddings live on a unit hypersphere; cosine similarity = dot product
- The towers never see other input — no cross-attention between image and text
  - Thus cheap: embeddings can be precomputed and indexed at inference time
  - In later fusion architectures, the models consider both image and text tokens
- Text encoder details: 63M params, width 512, 8 heads, BPE vocab 49152, max length 76, masked self-attention, features taken at [EOS]



Alignment goal



- Before training: encoders' outputs are not related (nothing relates v_dog to t_dog.) in common embedding space
- The training goal: pull each image toward its own caption, push it away from every other caption in reach — after which similar concepts also cluster (dog near cat, car near plane) as a side effect of shared captions and visual features.
- We never define “dog” anywhere. The structure comes entirely from co-occurrence: which texts appear with which images.

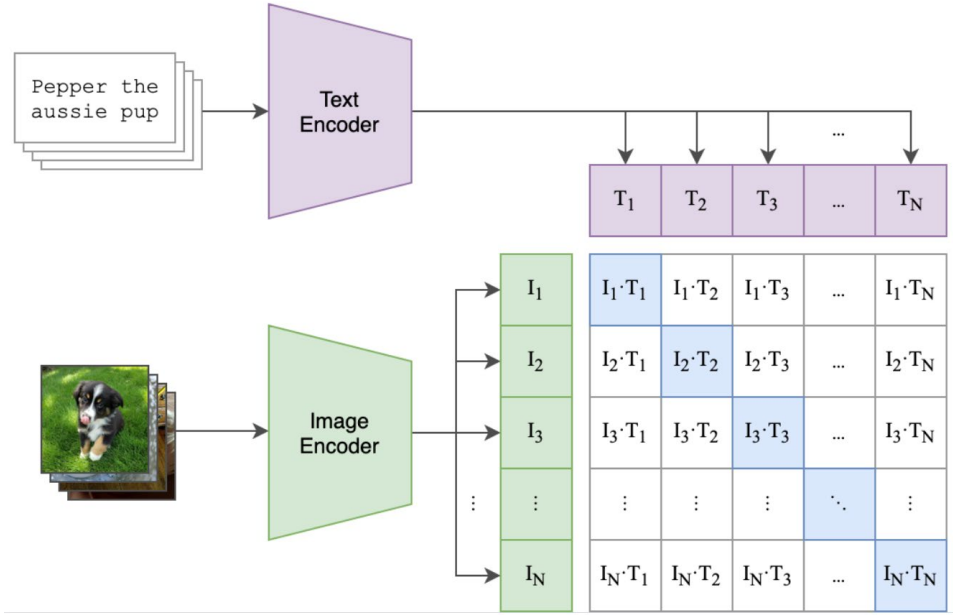
PART II

The contrastive objective

CLIP: scrape for image/text pairs, train contrastively

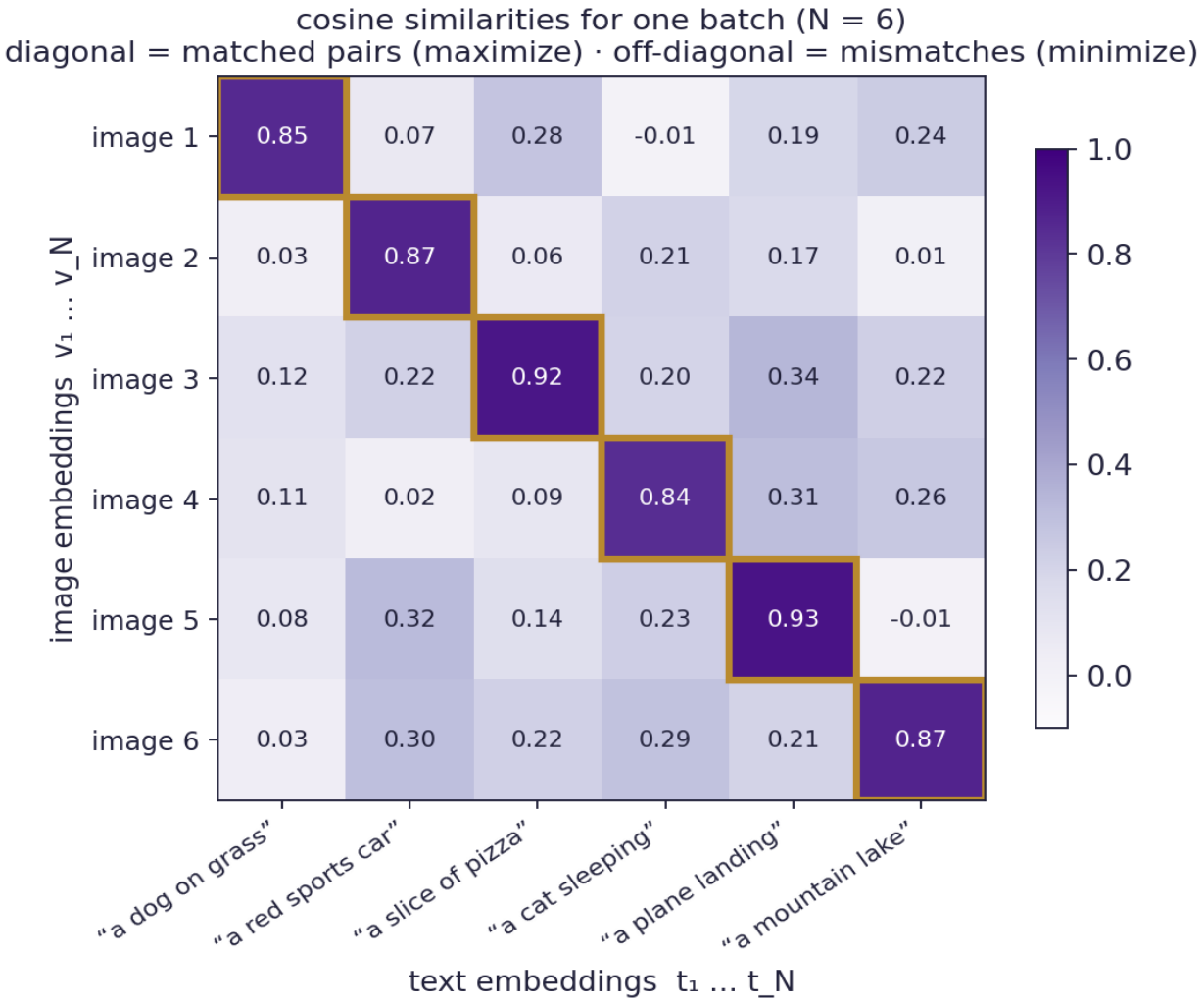
- Sample N (image, text) pairs from the web data: **$(I_1,T_1), (I_2,T_2), \ldots, (I_N,T_N)$**
- **No annotators required!**
- Within this batch we get, for free:
  - N positive pairs — (I_i, T_i): this caption accompanied this image (assumed correct/true)
  - $N^2 - N$ negative pairs — (I_i, T_j), $i \neq j$: almost certainly unrelated (with 400M pairs, collisions are rare)
- This is a self-labeling classification problem:

  for each image, "which of these N captions is yours?"
- Large batch size needed (why?):
  - CLIP trains with N = 32,768 → each positive is contrasted against 32,767 negatives
  - Approx $32768^2 \sim 10^9$ dot products per batch…very expensive
  - Benefit: more negatives we are forced to learn more complex associations between text and image embeddings



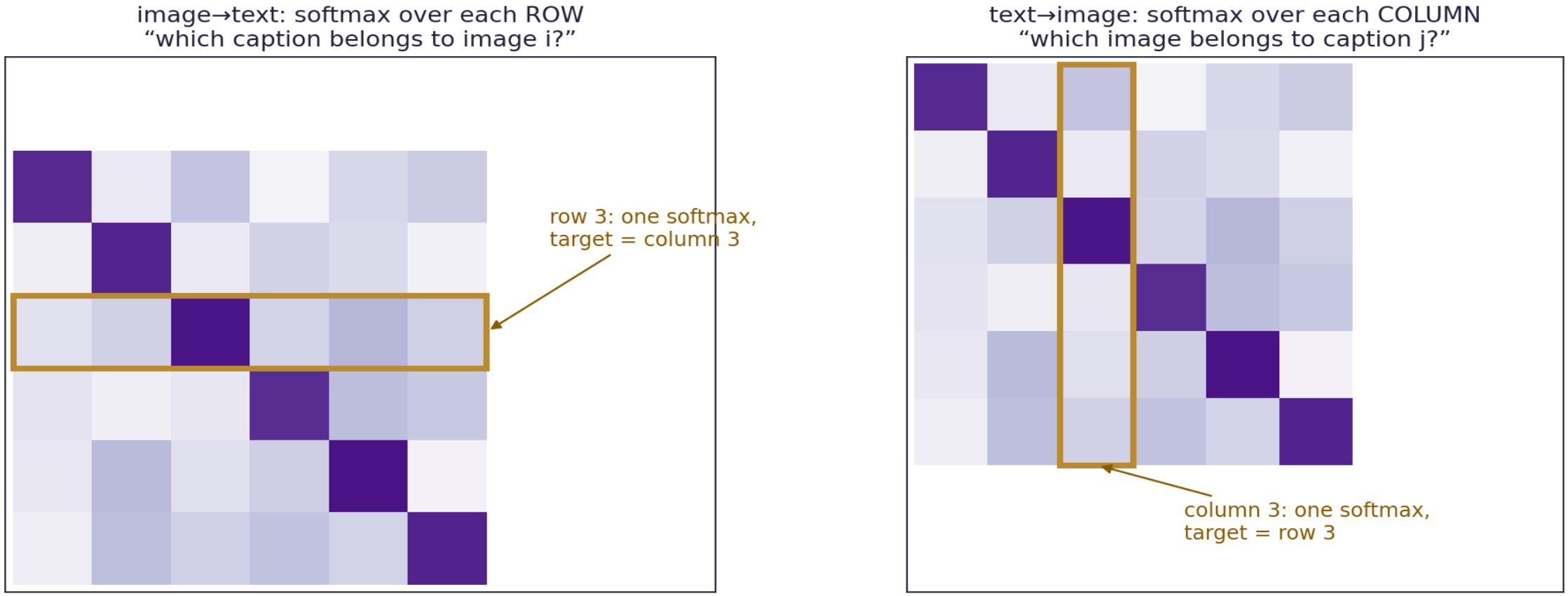
Radford et al. 2021, §2.3–2.5.

The N×N similarity matrix



- **Pack $v_1 \ldots v_N$ into matrix V and $t_1 \ldots t_N$ into matrix T**
- **All N^2 cosine similarities via 1 mx mult: $V T^t$**
  - Row i = image i scored against every caption
  - Column j = caption j scored against every image
- **Training objective: "maximize the diagonal elements and minimize everything else"**
- Note this "cosine matrix" is NOT symmetric
  - rows and columns answer different questions
    - Out of all these descriptions, which one fits this specific image best?
    - Out of all these images, which one represents this specific caption best
    - One to many association – Why?
  - Loss based on these values needs to deal with both query/match directions for generality.

Two classification/matching problems in one matrix



- Image→text: row i is treated as logits over N caption-classes; the correct “class” is i. One standard cross-entropy per row.
- Text→image: column j is logits over N image-classes; target j. One cross-entropy per column. Same matrix, inverted question — and both directions matter (we want retrieval to work both ways).

Symmetric InfoNCE loss

$$L_{img} = -(1/N) \sum_i \log \left[\exp(\cos(v_i, t_i)/\tau) \Big/ \sum_j \exp(\cos(v_i, t_j)/\tau) \right]$$

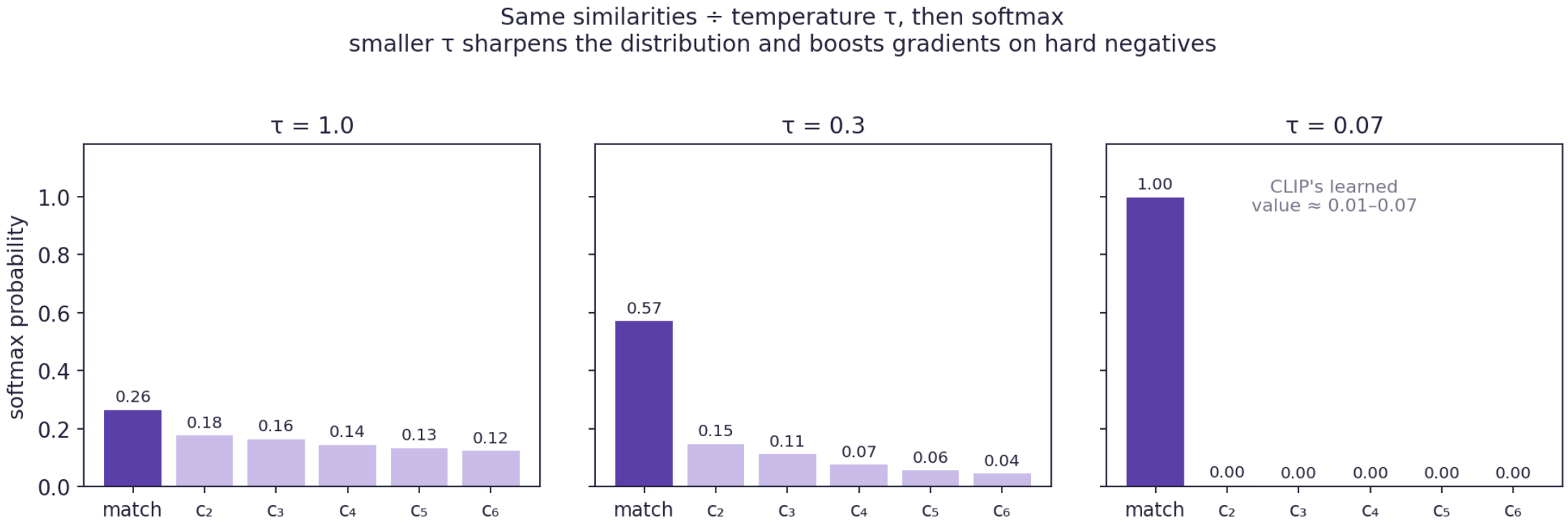
$$L_{txt} = -(1/N) \sum_j \log \left[\exp(\cos(v_j, t_j)/\tau) \Big/ \sum_i \exp(\cos(v_i, t_j)/\tau) \right]$$

$$L = (L_{img} + L_{txt}) / 2$$

- **In each line above: "cross-entropy loss" computed, where the logit for class j is (similarity/τ) and the target is the diagonal**
  - Numerator pulls matched pairs together; denominator pushes ALL mismatches in the batch apart
  - This is the InfoNCE family (van den Oord 2018): Information noise-contrastive loss
    - maximizing a lower bound on mutual information between image and text
  - Gradient intuition: the "hardest" negatives (highest wrong similarity) dominate the denominator's gradient — the model spends its capacity where it is most confused

Radford et al. 2021, §2.3; van den Oord et al. 2018 (InfoNCE, arXiv:1807.03748);

The role of temperature (τ)



- Cosine similarities occupy [−1, 1] — too flat for softmax to produce robust probabilities.
- Dividing by $\tau < 1$ rescales them ($\tau = 0.07$ → logits in [−14, 14]).
- Small τ → sharper/more compact distribution → higher penalties on hard negatives (but if too small → training instability).
- In CLIP: τ is learned (as "log-parameter", for stability), initialized at 0.07, clipped/capped so scaling ≤ 100

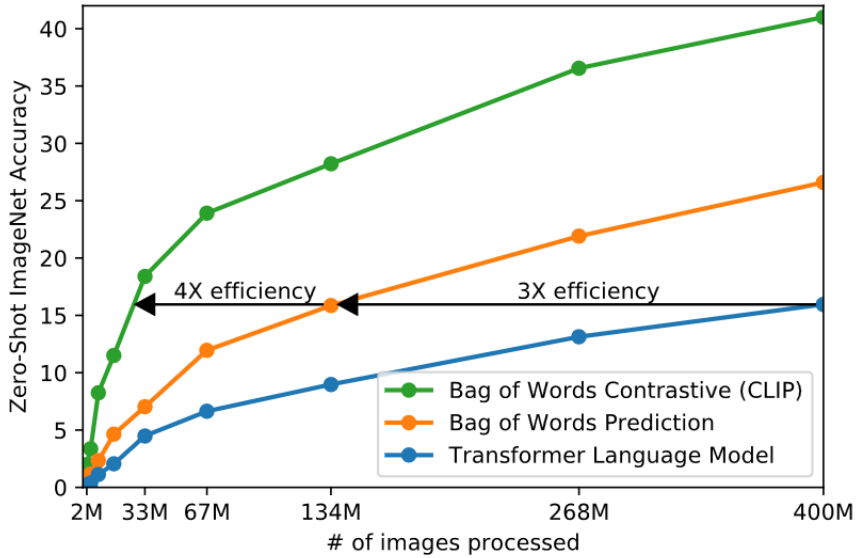
Core method ~ 10 lines of code

```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter
# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]
# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)
# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)
# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

- Algorithm flow: encoders → normalize into joint space → similarity matrix → diagonal labels → two cross-entropies → average.
- Everything else in the CLIP paper is about data, scaling and evaluation.

Question: Why not just predict the caption?

- **Alternative:** train an image-conditioned language model (a transformer) to generate the caption word-by-word
- The image is encoded to a feature and paired with text during training; at inference image is again encoded as input and LM autoregressively generates caption, word by word
- **Zero-shot learning efficiency on ImageNet:**
  - transformer LM predicting exact captions – the autogressive baseline was very slow and had poor predictive accuracy
  - Switching to predicting a bag-of-words of the caption (order dropped) — 3× more efficient
  - Switching to contrastive matching (CLIP) — a further 4× more efficient (≈12× vs the generative LM)
- **Why: exact wording is mostly not predictable from image**
  - "my dog Max at gran's, summer '19" — the visual content can't determine those words; the LM wastes capacity modeling "caption noise"
  - Matching simply asks: "is this text more compatible with this image than 32,767 others?"
  - Much less restrictive request: no worrying about grammar or subtleties around phrasing



Training CLIP: summary

Aspect	Value	Comment
data	WIT: 400M web (image, text) pairs, 500k query-balanced	scale + diversity replaces curation (curation not feasible at scale)
batch size	32,768	negatives come from the batch — bigger batch = solving harder task
image towers	Models investigated: 5 ResNets + 3 ViTs (B/32 → L/14@336)	ResNet + ViT; ViT up-scaling cut compute time
text tower	63M-param, 12-layer, width-512 transformer	Don’t need complex text encoding/processing
compute	up to 18 days × 592 V100s (RN50x64) / 12 days × 256 (ViT-L)	In 2021 “large”, but current VLM towers need vastly more compute
epochs	32 epochs over 400M pairs	≈ 13B image-text examples seen

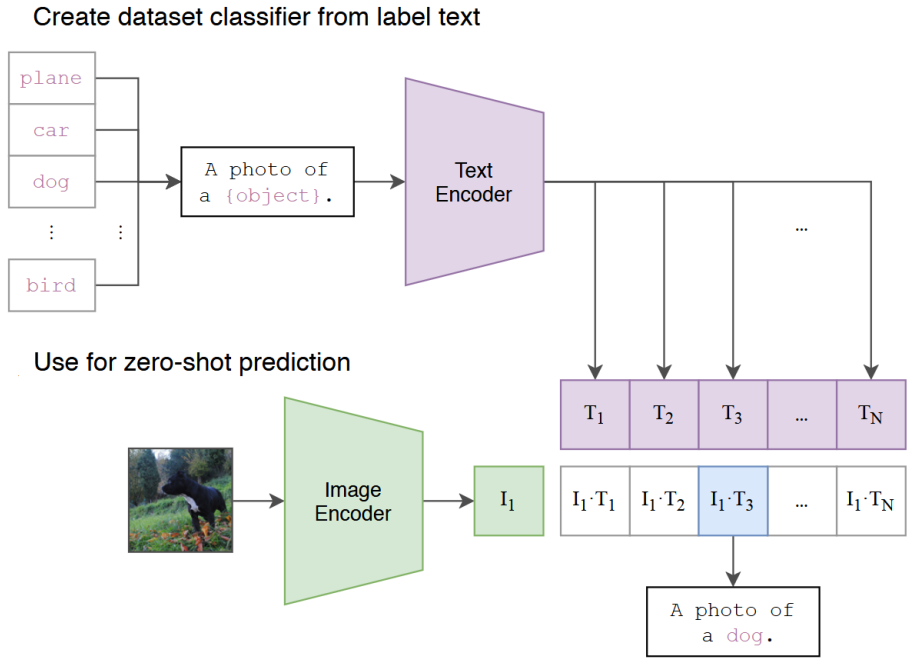
PART III

Zero-shot transfer

Image classification - Classes mapped to captions



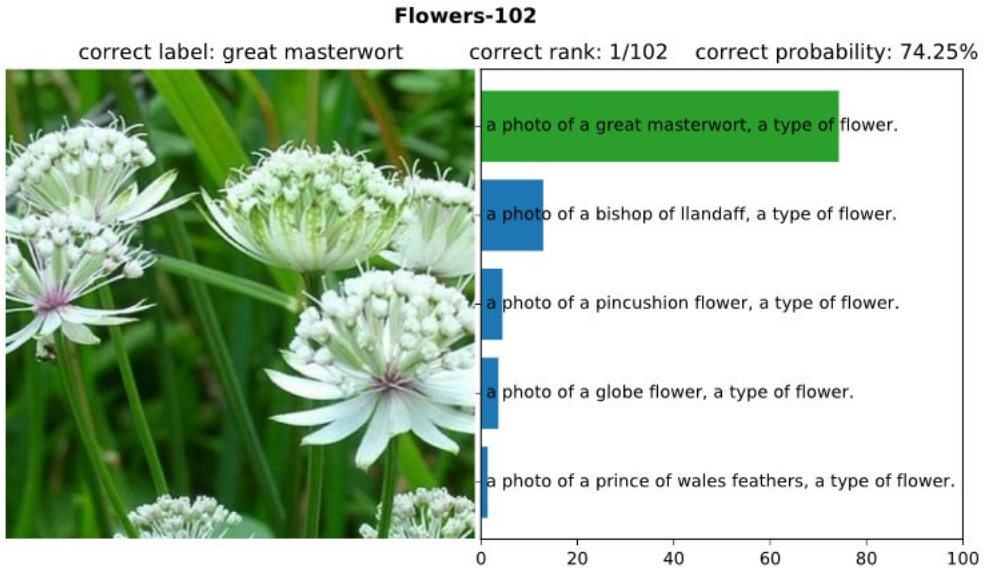
- **No training step, no gradient, no labeled examples:**
  - The text tower generates the classifier weights: each t_k is the embedding for the “class caption”
  - A new set of captions = a new classifier
  - Cost: embed query image, embed each target class caption, similarity, softmax, max
  - Why “zero-shot”? zero task-specific examples used — the knowledge came from 400M generic pairs in pretraining
  - Note: Zero-shot errors are usually near-misses in meaning (semantically similar), not random failure.



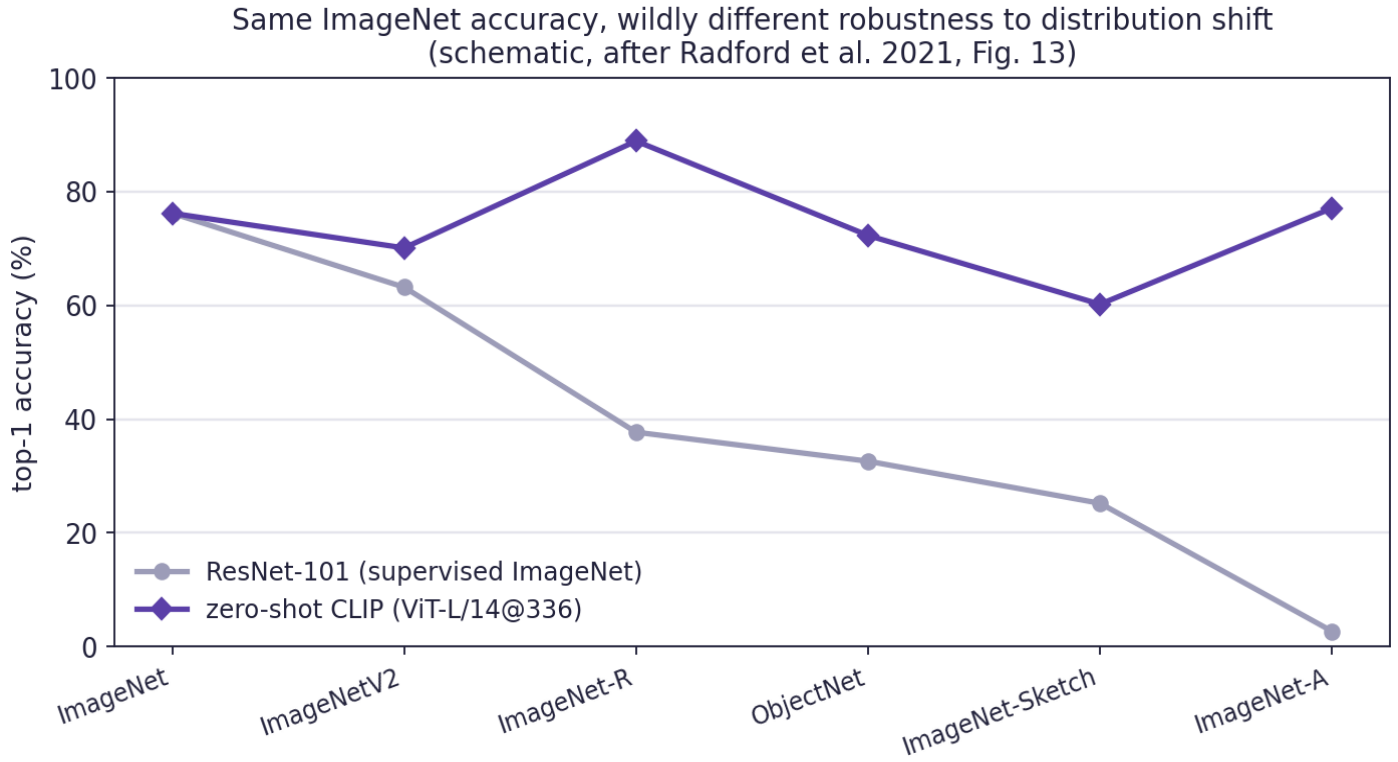
Prompt engineering before LLMs

- **The caption (“prompt”) you write determines the classification performance — wording matters**
  - Raw label “boxer” is ambiguous (dog? athlete?); training captions are full sentences, so single words are out-of-distribution
  - “A photo of a {label}.” → +1.3% ImageNet accuracy vs raw labels, from this phrasing alone
  - Domain templates help more: “a photo of a {label}, a type of pet.” (Oxford Pets) · “a satellite photo of {label}.” (EuroSAT)

- **Prompt ensembling**
  - Embed the same class through 80 templates (“a sketch of…”, “a close-up of…”, …) and average the text embeddings
  - Averaging happens in embedding space → classifier stays exactly one vector per class, zero inference-time cost
  - Prompt engineering + ensembling together: ≈ +5% ImageNet — “free” since model untouched.



How well does zero-shot work?



- **zero-shot CLIP ≈ 76.2% ImageNet top-1 — matching the original supervised ResNet-50, with NO ImageNet labels**
  - Across 27 datasets: wins on 16 vs a supervised ResNet-50 linear probe — strong on “natural” categories, weak on specialized/abstract tasks (satellite, tumors, counting)
- **Very robust (fig, left). The supervised models collapse under distribution shift (sketches, renditions, adversarial photos); CLIP barely moves**
  - Reason: it never overfits ImageNet's particular photo distribution — its “training distribution” was the whole web

- Caution for your projects: zero-shot ≠ universally good. Always check per-domain — CLIP's EuroSAT zero-shot was under 60% when a trained probe gets ~98%.

CLIP enables retrieval in both directions

- **Text→image search:** embed the query sentence once; nearest-neighbor over pre-computed image embeddings
  - Benefit of dual-encoder approach: the image database is indexed offline; queries cost one text-encoder pass + one k-NN lookup
  - Billion-image search engines run this pattern (they approximate NN over CLIP embeddings)
- **Image→text:** score an image against candidate captions — the same operation as zero-shot classification (a label set is just a tiny caption gallery)
- **Zero-shot classification, retrieval, deduplication, clustering: ALL reduce to nearest-neighbor in the joint space**
  - Useful Benchmarks: Flickr30k / MS-COCO recall@K — CLIP zero-shot rivals originally fine-tuned systems on Flickr30k

IMPORTANT: CLIP is not a classifier. It is a “Similarity Oracle” between images and sentences.

PART IV

Successors and limitations

ALIGN: scaling up massively, with no data curation

- To construct WIT a substantial amount of data cleanup/filtering was performed (costly)
- More image/text data should increase performance further
- ALIGN (A Large-scale ImaGe and Noisy-text embedding) addressed this with small changes
- Same basic approach as CLIP **dual encoder, contrastive, symmetric loss) — different bet on data**
  - dual encoder, contrastive, symmetric loss
  - BUT 1.8 BILLION (scaped) alt-text pairs, minimally processed (4.5x CLIP)
    - only frequency-based filtering, no query curation
  - Towers: EfficientNet-L2 (image) + BERT-Large (text)
- **Result: matches/exceeds CLIP on retrieval benchmarks despite far noisier text**
  - Profound impact: at billion scale, having enough data means noise is not very important — high quality curation is no longer a bottleneck for this task
- **Open-source projects: LAION-400M/5B datasets + OpenCLIP reproduced and extended this publicly — for all to use**

Jia et al. 2021, “Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision” (ALIGN, arXiv:2102.05918); Schuhmann et al. 2022 (LAION-5B).

SigLIP: replace the softmax with a sigmoid, simplify loss

- CLIP's softmax compares every candidate image to the whole batch (the denominator sums over all captions)
- The CLIP softmax loss changes with batch size, and very large batches need expensive “all-gather” distributed GPU communication
- SigLIP change: treat every cell (i, j) of the matrix as its own independent binary problem — “matched or not?”
- Uses sigmoid **σ()** to rescale logits independently and introduces a bias **b** (also learned)
- Bias avoid initial training issues caused by massive dominance of negative examples (N^2-N negatives vs N positives) which are independently considered; high initial value ensures the model accommodates many negative examples without destabilizing

$$L = -(1/N) \sum_i \sum_j \log \sigma(z_{ij} \cdot (t \cdot \cos(v_i, t_j) + b)) \qquad z_{ij} = +1 \text{ if } i=j, \text{ else } -1$$

t, b learned (b init ≈ −10: with N−1 negatives per row, start pessimistic to avoid a wall of early false-positive gradient)

- **What this gives us:**
  - No cross-example normalization → loss is stable across batch sizes; simpler multi-device training (no logits all-gather)
  - Strong at moderate batches (performance saturates ~32k — at smaller batches still good, so faster training)
  - **SigLIP towers are the default vision encoder in many current open VLMs**

Zhai et al. 2023, “Sigmoid Loss for Language Image Pre-Training” (arXiv:2303.15343).

CLIP vs SigLIP

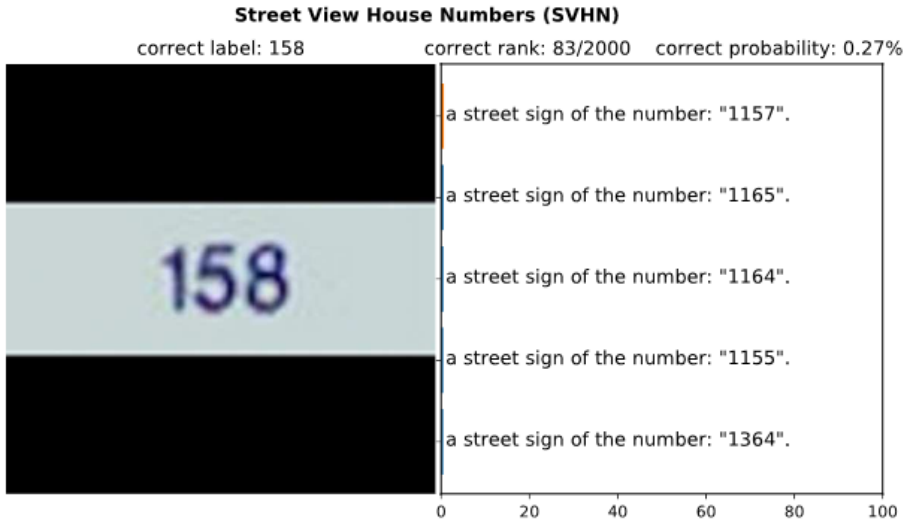
	CLIP (softmax InfoNCE)	SigLIP (sigmoid)
unit of loss	one softmax per row + per column	one binary term per matrix element
normalization	across the whole batch (denominator)	none — each pair independent
batch-size dependence	strong: negatives = batch − 1; quality scales with N	weak: works well from ~4k; saturates ~32k
distributed cost	all-gather of embeddings/logits across devices	pairwise chunks can be streamed device-to-device
extra parameters	temperature t (learned)	temperature t + bias b (b init ≈ −10)
class imbalance	hidden inside softmax (“automatically” dealt with)	explicit: N positives vs N^2−N negatives → hence need bias to damp

- Same geometric relationships learned for both — a shared image-text embedding space.
- Both losses target “is this pair matched?” but differ in what each answer is normalized against.

Zhai et al. 2023, §2–3.

Known failure modes

- **1 · Bag-of-words behavior**
  - "a man in a red shirt next to a blue car" vs "a man in a blue shirt next to a red car" — CLIP often scores these nearly identically (word order barely changes the embedding)
  - Benchmarked by ARO/Winoground: vision-language models only near chance on relation/order tests, but excel at retrieval
  - Cause: contrastive training seldom needs composition/order — nearest-neighbor over web pair embeddings is often enough
- **2 · Counting and fine spatial relations**
  - "three dogs" vs "four dogs", "left of" vs "right of" — as above
- **3 · Anything requiring more output tokens to be considered**
  - The whole image is compressed to 512 numbers
  - But dense tasks (localization, OCR, multi-step reasoning) need the patch tokens CLIP throws away (we look at using this info in later lecture)



Yuksekgonul et al. 2023 (ARO, arXiv:2210.01936); Thrush et al. 2022 (Winoground, arXiv:2204.03162); Radford et al. 2021 §6 (limitations).

ClIP Summary

Alignment	Two independent towers (ViT + text transformer) project into one L2-normalized space where cosine similarity means semantic match. No cross-attention needed (cheap).
Loss function	The batch labels: N×N similarities, diagonal targets, symmetric cross-entropy with learned temperature (InfoNCE).
Zero-shot	The text tower outputs classifier embeddings based on class “captions”, no training/tuning, low compute – simple cosine similarity, robust to domain shift
Limitations	One vector per image → bag-of-words limitations, weak counting, no “dense tasks”. SigLIP fixes the stability and training efficiency (uses sigmoid not softmax, batches “decoupled”);

Resources

- **References (for information)**
  - Radford et al. (2021). Learning Transferable Visual Models From Natural Language Supervision (CLIP). ICML 2021. arXiv:2103.00020
  - Jia et al. (2021). Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision (ALIGN). ICML 2021. arXiv:2102.05918
  - Zhai et al. (2023). Sigmoid Loss for Language Image Pre-Training (SigLIP). ICCV 2023. arXiv:2303.15343
  - van den Oord et al. (2018). Representation Learning with Contrastive Predictive Coding (InfoNCE). arXiv:1807.03748
  - Yuksekgonul et al. (2023). When and why VLMs behave like bags-of-words (ARO). arXiv:2210.01936 · Thrush et al. (2022). Winoground. arXiv:2204.03162
- **Online**
  - HuggingFace CV course, multimodal unit: huggingface.co/learn/computer-vision-course
  - https://huggingface.co/learn/computer-vision-course/unit4/multimodal-models/clip-and-relatives/clip